Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?Answer Choices:A. It activates TLR5 on the surface of macrophages.B. It primarily induces the Th1-cell response.C. Peptidoglycan is its major antigen that induces an immune response.D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient is diagnosed with community-acquired pneumonia, which is typically caused by bacteria such as Streptococcus pneumoniae, Haemophilus influenzae, or Moraxella catarrhalis. The immune response elicited by these pathogens involves the recognition of peptidoglycan, a component of the bacterial cell wall, by the immune system. Peptidoglycan is a major antigen that induces an immune response, leading to the activation of macrophages and the production of cytokines. This response is typically associated with a Th1-cell response, which is characterized by the activation of macrophages and the production of interferon-gamma. The pathogen does not evade the immune response by encapsulation, as encapsulation is a characteristic of encapsulated bacteria such as Streptococcus pneumoniae, which is still recognized by the immune system. Toll-like receptor 5 (TLR5) is primarily involved in the recognition of bacterial flagellin, which is not the primary component of the pathogens causing community-acquired pneumonia. </think><answer>C. Peptidoglycan is its major antigen that induces an immune response. </answer>